

## Agenda

- What is a Schema Design
- How to Approach Schema Design
- Representing Relationships : Cardinality
- Sparse Relationships

## What is Schema Design

Structure of database  
Tables in Database  
Columns in Table  
Key - Primary Key, foreign Key  
Index

Pictorial representation of DB  
How it is structured

# How to Approach Schema Design

eg Build a software for Scaler

## Requirements

- ① Scaler <sup>DB</sup> will have multiple Batches <sup>Table</sup>
- ② for each batch, we need to store name, startMonth, current Instructor <sup>cols</sup> <sup>Table</sup> <sup>Table</sup>
- ③ for Batch, there can be Multiple Students
- ④ for Batch, there will be multiple Classes
- ⑤ for each class store name, date & time <sup>Table</sup>  
of class <sup>cols</sup>
- ⑥ for every student, Name, grad year, University, email and Phone Number <sup>cols</sup>
- ⑦ Every Student has a Buddy who is also a Student <sup>PK</sup>
- ⑧ Student can move from one batch to another
- ⑨ for every student there is Mentor <sup>Table</sup>

- ⑩ For every mentor, name & Company
- ⑪ For every Mentor Session <sup>Table</sup> 
<sup>cols</sup>  
 (time, duration  
 Student, Mentor  
 Student Rating  
 Mentor Rating  
<sup>cols</sup>
- ⑫ For every Batch, store whether it is  
an academy batch or DS ML Batch  
Type Col

find all Nouns they will represent entities  
 or cols of entities

Batches  
 Instructor  
 Students  
 Classes  
 Mentors  
 Mentor Session

Cols in table

→ Primitive → Direct info Basic Data Type  
 → Relationship → Students → Batch

## Primary key

Identifier that DB generates automatically

[ <Table Name> - id ]  
Student - id

## Batches

batch-id, Name, Start-Month

## Students

Student-id, name, email, phone-number, grad-year  
univ\_name

## Instructors

Inst-id, name, email, avg-rating

## Classes

Class-id, name, date, time

## Tutors

Mentor id, name, company-name

MentorSession

MentorSession id, time, duration, student-rating  
mentor-rating

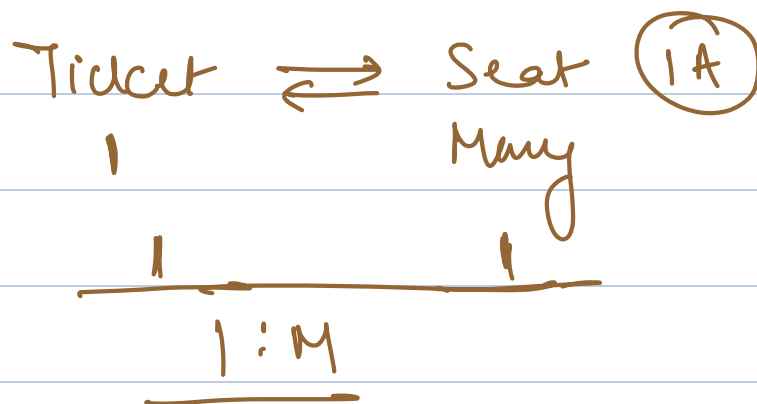
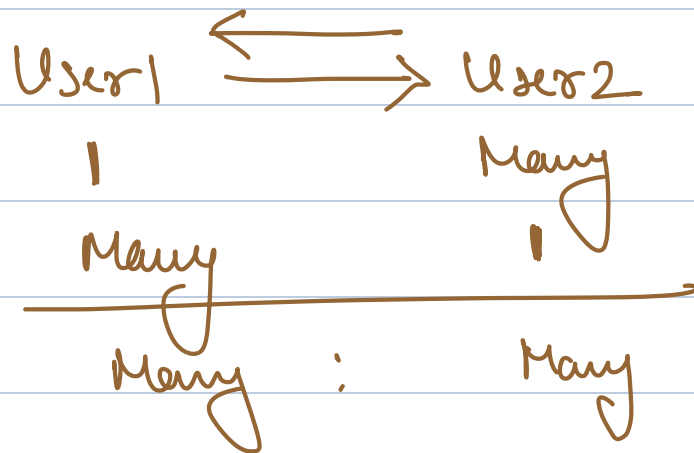
## CARDINALITY

How one entity is associated with another entity

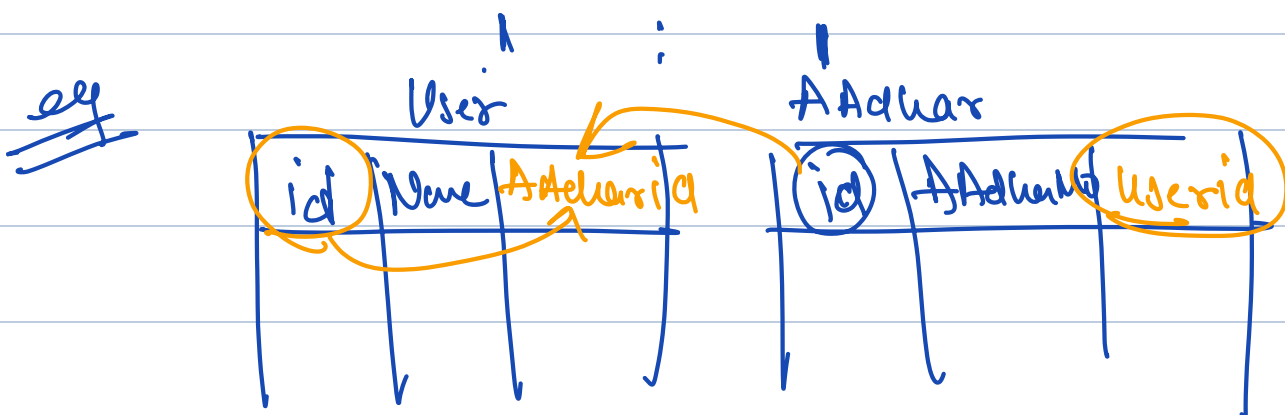
- 1 Parent can have Many children
- 1 Person can have Many Cars
- 1 Aadhar can belong 1 Person

|           |                                      |
|-----------|--------------------------------------|
| 1:1       | one entity is related one entity     |
| 1:M   M:1 | one entity is related Many entities  |
| M:M       | Many entity is related Many entities |

# How to identify Cardinality



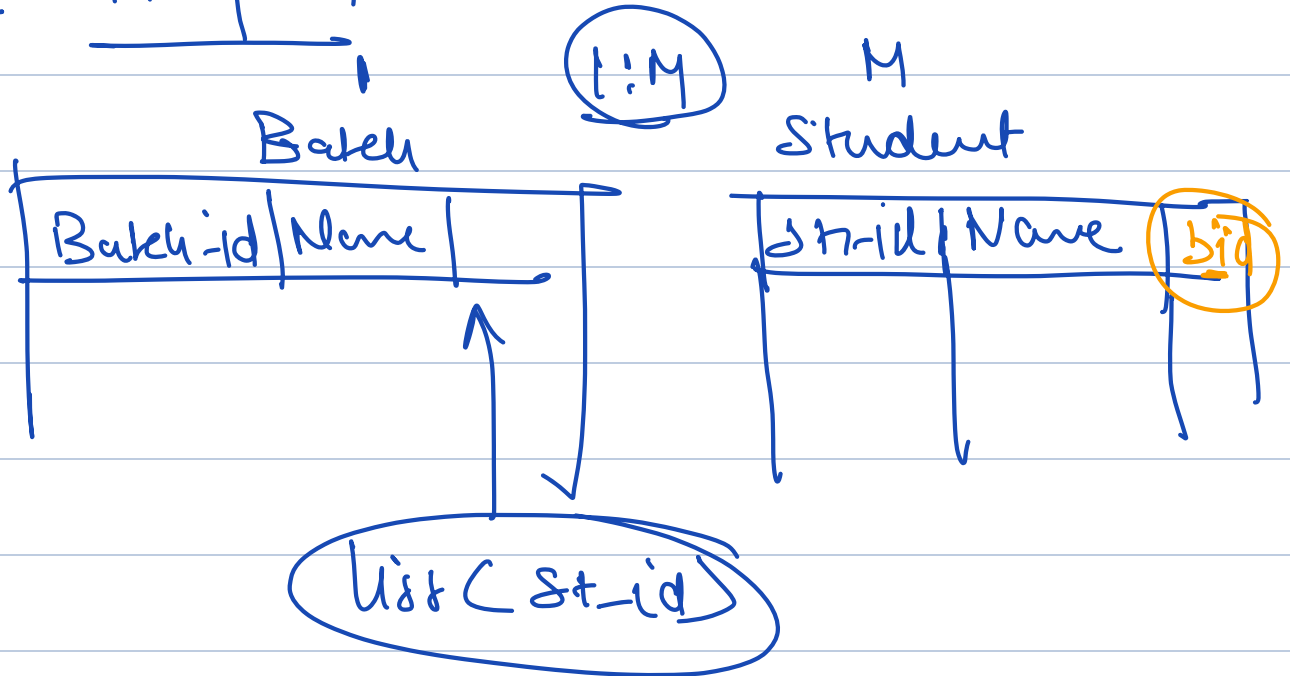
## Represent Relation in Tables



Id of 1 side on another side

Q4

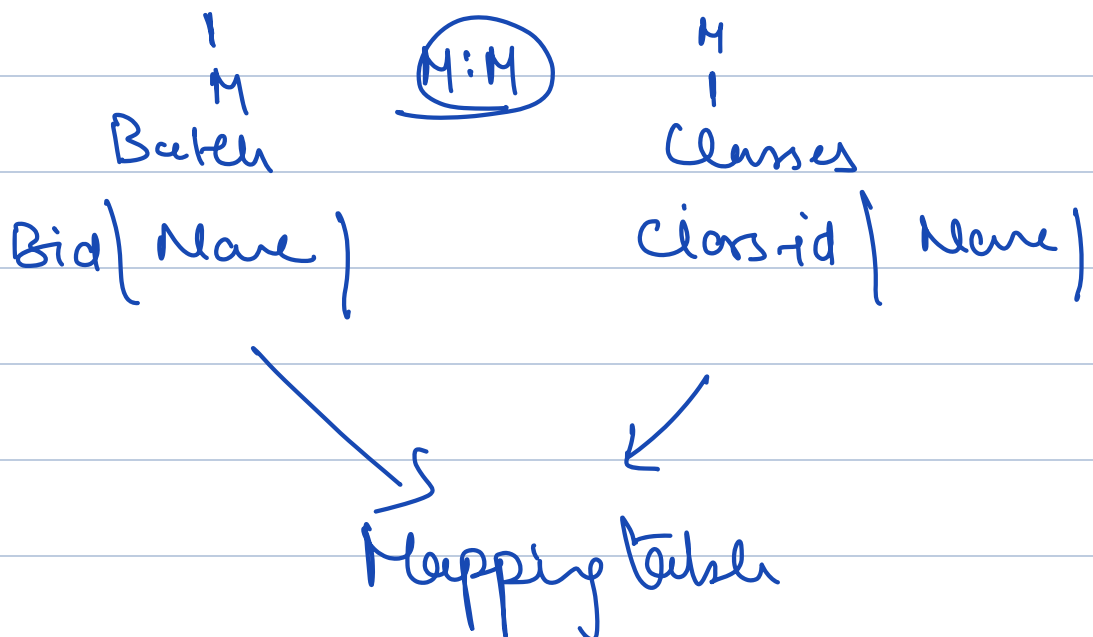
1:M / M:1



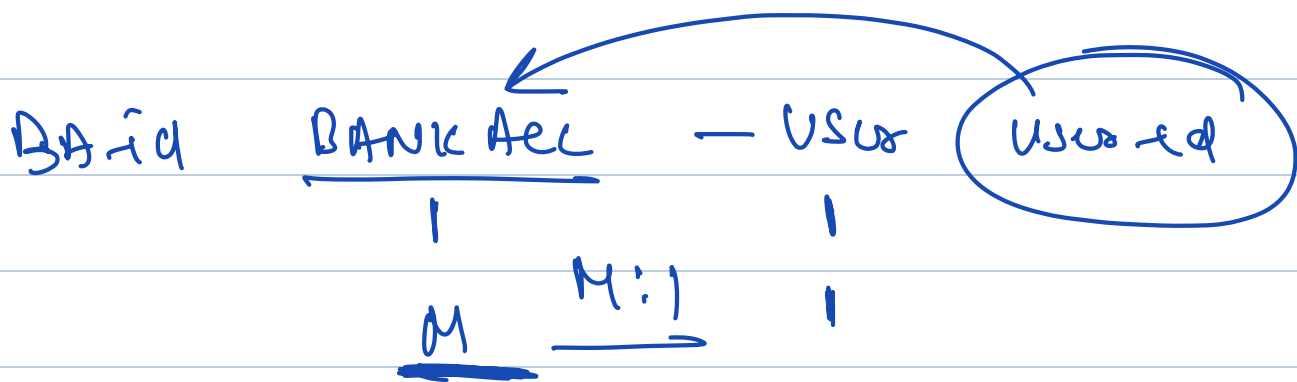
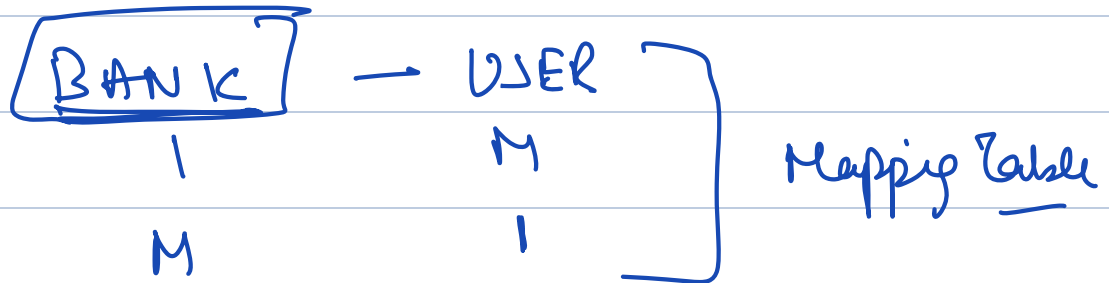
Id of 1 side on the Many side

③

M:M



| B-id | Class id |
|------|----------|
| .    | .        |
|      |          |



## ENUM

Bank Type → Academy ] string  
DSML

| Bid | Name | Type (string) |
|-----|------|---------------|
| 1   | B1   | DSML          |
| 2   | B2   | Academy       |
| 3   | B3   | Academy       |
| 4   | B4   | DSML          |
|     |      |               |

Type  
 (Academy)

Slower Comparison  
Take more space

Use Integer

0 - DSQL

~~1 - Academy~~

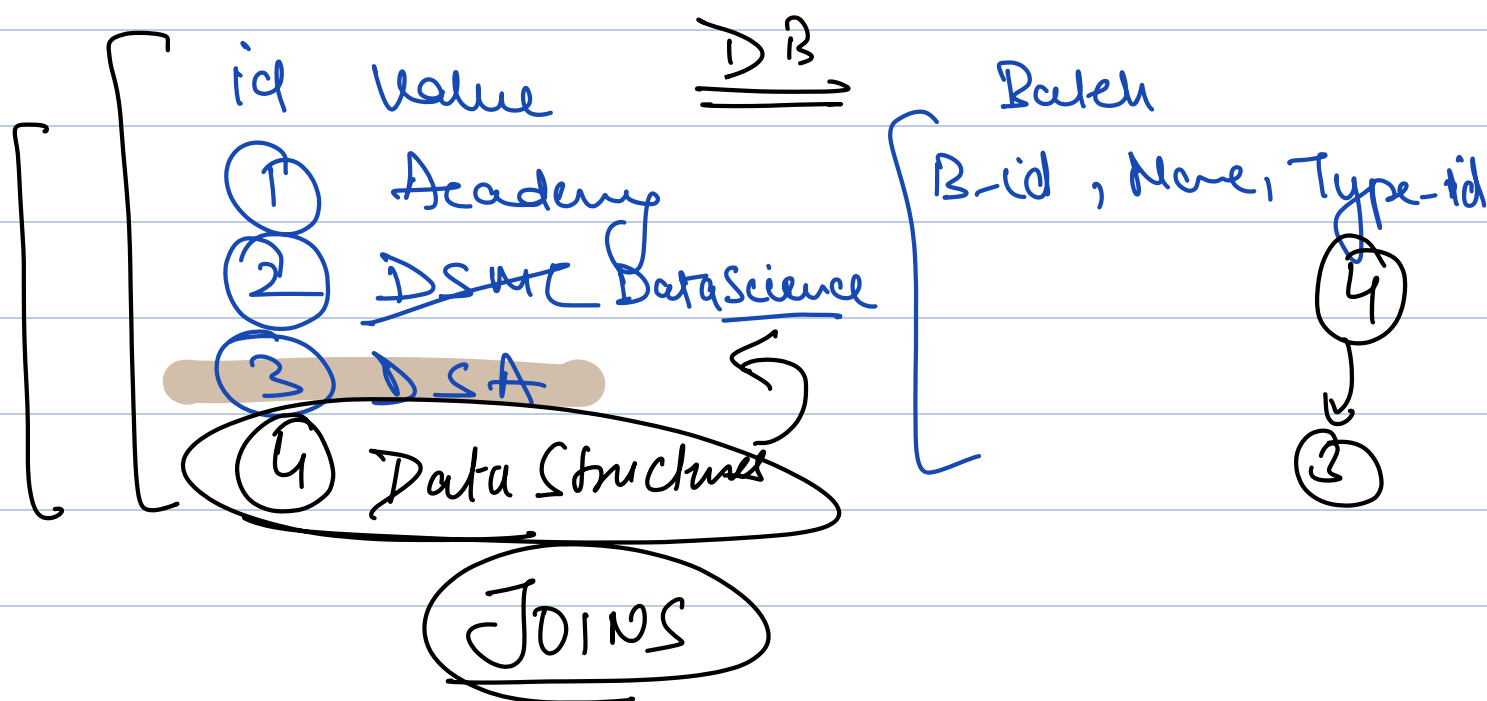
1 - DSA

~~3 - DSA~~

Code

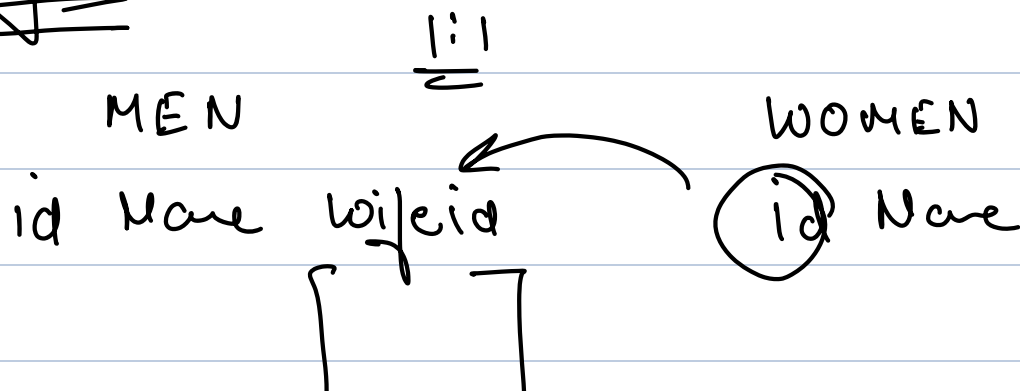
3 - Data Structures

LOOKUP Table



Sparse Relationship

City DB



1M Men

1M Women

↳ 20000 Married

NULL

Sparse Relationship

↳ Look UP Table

| men id | women id |
|--------|----------|
|        |          |

30K